

MITCHELL GRAY

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in MitchellEGray

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Education

Georgia Tech, Atlanta, GA

Master of Science, Computer Science

GPA: 4.0

Artificial Intelligence

Cornell University, Ithaca, NY

Bachelor of Science, Computer Science

Distributed Systems

Minor, Operations Research Information Engineering

M.Eng Equivalence

Focus Areas: Distributed Systems · Systems Programming · AI · Machine Learning · Databases · Cloud

Technical Skills

Languages: C · C++ · Python · Java · Rust · PL/SQL

Software & Tools: Git · Pytorch · Docker · Kubernetes · Scikit-learn · NumPy · Pandas · Poetry · CMake · CI/CD

Relevant Work Experience

Oracle

Redwood City, CA

Software Engineer II | C, PL/SQL, Raft, Java, OracleDB, OCI

Sept 2025 – Present

- Developing Raft Replication infrastructure in the Oracle Globally Distributed Database used for Shard Replication · Writing fault-tolerant and scalable Distributed Systems code in **C**, **Java**, and **PL/SQL**.
- Received **Training & Mentorship Excellence Award** · Provided technical mentorship to our newest engineer.
- Designed and implemented a generic shard-operations marker framework that consolidated multiple ad-hoc code paths into a single framework · Improved maintainability, scalability, and ease of creating new markers with fewer defects.
- Architected and implemented advanced deployment algorithms when specifying server rack requirements to more evenly balance replicas across shards automatically · Improved CLI user-interface to view user-specified rack information.
- Implemented comprehensive Pluggable DBs lifecycle management · Enabled seamless Drop, Unplug, Plug, Clone, and Relocate operations to streamline maintenance for Raft replicas using Oracle's multi-tenant container architecture.
- Led remediation of shard-to-shard TDE encryption defects by unifying divergent serialization paths · Preserved encryption metadata during replica transfer, enabling reliable decryption on receiving shards and restoring correctness for TDE-enabled shard replication workflows.
- Improved slow-replica flow control · Implemented new throttling logic to stabilize replication with degraded replicas.

Software Engineer I

June 2024 – Sept 2025

- Engineered a fault-tolerant replica replication feature on Oracle Cloud Infrastructure · Enforced placement of Raft replicas on separate server racks to eliminate single-rack replication-unit failures and improve database availability.
- Refactored over 1,200 lines of Sharding serialization code to eliminate a critical data-loss defect · Designed robust de/serialization pipelines and multi-format conversion logic that greatly improved data integrity and system reliability.
- Optimized multiple replica-to-replica data transfer operations by using batching · Achieved a **700%** performance improvement offline and **250%** improvement online after optimization code changes.
- Resolved critical recoverability issue in replica-to-replica data transfer operations by ensuring complete transfer of essential recovery-related tables and data between replicas · Improved reliability and system integrity during failure.
- Enhanced error-handling mechanisms throughout the codebase · Improved error handling reliability and diagnosability.

Gecko Robotics

Pittsburgh, PA

Software Engineer Intern | C++, Python, C, CMake, GCP, CI/CD

May 2023 – Aug 2023

- Worked on the Robot Controls team · Wrote code for an asynchronous distributed system in **C++**, **Python**, and **C**.
- Revamped Robot & DAQ emulators · Implemented communications protocol and Client/Server TCP networking.
- Replaced VS build-system with **CMake** · Added calibration support to emulators · Integrated **CI/CD** and **Poetry**.

Carnegie Mellon University - Software Engineering Institute

Pittsburgh, PA

DevOps Engineer Intern | Python, Rust, CI/CD, Docker, Kubernetes

May 2022 – May 2023

- Developed **Python**, **Rust**, and **Bash** code on various NDA projects · Updated and created **CI/CD** pipelines · Created GitLab **REST API** live data visualizations for clients · Used ArgoCD and Helm to deploy **Kubernetes** clusters.
- Built internal analytics tools using **Python**, **Neo4j**, NeoDash, and PageRank-based graph analysis to generate engineering metrics and documentation adopted across multiple teams within the company.

Relevant Academic Work Experience

Georgia Tech Contextual Computing Group

Atlanta, GA

Graduate Researcher | AI, ML, Python, Typescript

June 2025 – July 2026

- Worked on Nosi, an IDE that combines an encrypted virtual file system and application behavior-logging · Utilizes **Machine Learning** techniques to analyze behavioral data and flag potentially-cheating students.
- Published at **Learning at Scale 2026** · Integrated the IDE into multiple Georgia Tech graduate-level CS courses.
- Created models that can detect the difference between human-coding and code transcription · Implemented tools to help review flagged cheating cases and replay student interactions with the IDE · Developed scripts that automatically generate encrypted repositories for students and decrypt them for the auto-grader.

Georgia Tech

Atlanta, GA

Graduate Teaching Assistant | AI, Python, NumPy

May 2025 – May 2026

- Taught graduate-level **Artificial Intelligence** · Helped with new plagiarism-detection tooling integration.

Cornell University

Ithaca, NY

Undergraduate Teaching Assistant | Python, NumPy, SQL, Java

Aug 2022 – May 2024

- Received the **Course Staff Exceptional Service Award** · Taught Databases, Object-Oriented Programming, and Physics · Lectured 400+ students with custom-designed lecture content · Led Exam sessions and other TAs.

Cornell Database Group

Ithaca, NY

Undergraduate Researcher | ML, Java, AWS

Jan 2022 – May 2024

- Published and presented paper at **VLDB 2023** combining **Reinforcement Learning** and Worst-Case Optimal Joins.
- Worked on upgrading query engine to a **Distributed Architecture** · Created dynamic statistics & RL visualizations.

Publications

Learning at Scale (2026)

[<https://dl.acm.org/doi/abs/10.1145/3774398.3811574>]

Nosi IDE: Securing Coding Assessment Integrity in the Age of LLMs via Process-Driven Analytics

Learning at Scale (2026)

[<https://dl.acm.org/doi/abs/10.1145/3774398.3811574>]

Demonstrating Nosi IDE: Securing Coding Assessment Integrity in the Age of LLMs via Process-Driven Analytics

VLDB (2023)

[<https://dl.acm.org/doi/abs/10.14778/3611540.3611629>]

Demonstrating ADOPT: Adaptively Optimizing Attribute Orders for Worst-Case Optimal Joins via Reinforcement Learning

Awards

Oracle, Database Organization

Mentorship and Leadership Excellence Award

Recognizes outstanding efforts in mentorship and training of others that go above and beyond the call of duty. Multi-day, multi-recipient efforts beyond the direct organization.

Cornell Bowers CIS

Course Staff Exceptional Service Award

Faculty-nominated award for exceptional commitment, hard-work, dedication, and exceptional service of a Teaching Assistant during the 2023-2024 academic year.

Relevant Projects

Inference Serving Platform | Personal Project | *Go, Kubernetes, vLLM, Docker* | *Github*

April 2026 – Present

- Building a cloud-hosted inference serving platform for deploying open-source LLMs behind scalable HTTP APIs · Implementing request routing, model container orchestration, autoscaling, and GPU-aware scheduling on Kubernetes.
- Integrating vLLM-based inference workers with asynchronous job queues, health checks, structured logging, and deployment metadata to support reliable model serving across multiple replicas.

Improving 3D Point-Cloud Classification | Deep Learning | *AI, Python, PyTorch* | *Github*

Aug 2025 – Dec 2025

- Re-implemented and upgraded state-of-the-art point cloud classification Deep Neural Networks · Implemented PointNet and PointPillars · Upgraded these models by modifying their architectures, adding Dropout, implementing Data augmentation, and introducing Residual Connections · Improved validation accuracy by 4% and massively reduced overfitting.

Distributed Testing Platform (SPEED) | Masters Project | *Java* | *Github* | *BOOM*

Aug 2023 – May 2024

- 1 of 44 projects selected for BOOM 2024 · Worked on SPEED, Scalable Platform for Efficient Execution of Distributed testing, which is a fault-tolerant system that contains a controller node that orchestrates worker nodes running **JUnit** tests on **Java** code · Aggregated worker node findings to the controller · Displayed test results in the frontend to the user once all tests finish.

SpGEMM Parallelization on GPUs | Parallel Computing | *CUDA, C++* | *Github*

Jan 2024 – May 2024

- Parallelized sparse-matrix multiplication operations using GPUs · Experimented with multiple variations of the algorithm which massively improved performance of the algorithms.